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A hybrid approach of entropy machine learning and TOPSIS method to evaluate financial performance of firms

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Abstract: *The objective of this study is to propose a hybrid approach of entropy machine learning and TOPSIS to rank the performance of different alternative firms for a long-term investor to select best firms for the portfolio. We used entropy machine learning method to determine the weights of selection criteria based the financial performance of firms and TOPSIS method to determine the ranking for each alternative firm based on ideal minimum (maximum) distance for the best (worst) possible criteria. Most of the studies have limitation of using equal weights of selection criteria. This study addresses this limitation by employing the entropy approach of weight determination. The sample size consisted of 10 firms from Refinery, Oil and Gas Marketing, and Oil and Gas Exploration sectors of Pakistan stock exchange and the financial data consisted of six years from 2019 to 2024. This study contributes to literate by addressing the limitation of entropy techniques containing negatives values by using more powerful software (MATLAB) and complex numbers for weight determination. The results of the ranking of firms were compared with the annual stock returns of the firms. The results indicate that, in most cases, the ranks of our method are very close to the ranks based on annual stock returns of firms. This study is useful for individuals and institutional investors to help them select best alternatives for their portfolio selection based on financial performance of companies.*

Key Words: *Entropy, Machine learning, TOPSIS, MATLAB, Portfolio selection technique, portfolio management, financial performance, MCDM*

1. INTRODUCTION

One of the major research avenues of investment decisions is portfolio management. While selecting a stock for a portfolio, there are several factors a portfolio manager must consider. These factors may broadly be classified into two as fundamental and technical factors. The technical analysis uses price trends to predict the prices and returns whereas the fundamental analysis uses analysis key financial and market variables of a firm. This paper analyzes the efficiency of selected firms in Refinery, Oil and Gas Marketing, and Oil and Gas Exploration sectors of Pakistan stock exchange using a two-stage blend of The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method and entropy method of muti-criteria decision making analysis method. Although various time series econometric studies are in abundant, the literature reports a very little number of studies for applications of TOPSIS methodology in stock investment decisions. The TOPSIS method was proposed by Hwang and Yoon in 1981 (Chen & Tsao, 2008). They introduced the concept of compromised solution. The compromised solution is choosing the best alternative nearest to the positive ideal solution and at the same time farthest from the negative ideal solution. The closeness in TOPSIS is measured using minimum Euclidean distance.

The fundamental analysts usually use traditional financial ratios to compare several firms. Hosaka (2024) argues that this traditional method is insufficient method to measure the performance and suggesting using several multi-lateral ways. The literature reports several studies that follow the methodology of multi-criteria decision making to assess the performance of firm using financial ratios. Most of the studies such as Wanke et al. (2016) and Bulgurcu, (2018) use equal weight for several selection criteria which is very impractical in reality. All the financial ratios have different relative significance level. For example, the decrease is sales turnover will not have a greater impact on the profitability as compared to same level of change in debt-to-equity ratio. Each ratio has a different relative significance therefore each ratio should have a different weight while selecting a best stock for the portfolio.

The entropy approach initially invented by Shannon (1948) is a machine learning approach that uses probability theory and randomness to measure disorder of information being processed in machine learning. The entropy measures the degree of information loss or impurity in the system. In this study we use entropy to obtain the weights of selection criteria. Therefore, this study contributes in the literature in various aspects. Firstly, unlike several studies, this study uses entropy to calculate the weights of selection criteria. Secondly, the computing power of entropy has been addressed specifically for negative ratios using the complex numbers. Thirdly, this study specifies and justifies the suitability of TOPSIS methodology by comparing its results with the actual annual stock returns of the sample firms.

1.1 Research Objectives

1. To develop a hybrid decision-making model using entropy-based machine learning and the TOPSIS

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method for ranking firms based on financial performance.

2. To determine the objective weights of financial performance criteria using the entropy method.

3. To apply the TOPSIS method to rank firms in the Refinery, Oil and Gas Marketing, and Oil and Gas Exploration sectors of the Pakistan Stock Exchange.

4. To compare the rankings derived from the proposed model with the actual annual stock returns of firms.

5. To demonstrate the suitability of the proposed model for aiding individual and institutional investors in portfolio selection.

1.2 Research Questions

1. How can a hybrid model combining entropy and TOPSIS improve the ranking of firms for portfolio selection?

2. Can the entropy method provide more realistic and objective weights for financial performance indicators compared to equal weighting?

3. How consistent are the firm rankings generated by the proposed hybrid model with the actual annual stock returns?

4. Does the use of complex numbers and MATLAB enhance the capability of entropy-based weighting for handling negative financial values?

5. Can this hybrid approach serve as a reliable tool for long-term investment decision-making in the energy sector?

1.3 Research Significance

This study contributes to the growing literature on data-driven investment analysis in several ways. It bridges the methodological gap by introducing a hybrid entropy-TOPSIS framework for firm evaluation, enhancing the robustness of portfolio selection strategies. It introduces a more nuanced weighting system for financial ratios, reflecting their actual influence on firm performance rather than assuming uniform importance. The research offers practical value to investors and portfolio managers by comparing model-based rankings with real-world returns, thereby validating the effectiveness of the approach. For academics and analysts, the study provides a replicable and adaptable evaluation framework for use in different markets or sectors.

1.4 Research Gap

Although technical and fundamental analyses are commonly used in stock evaluation, most fundamental analysis studies rely on traditional ratio comparisons without applying systematic weighting methods. Moreover, limited literature exists on the use of entropy-weighted TOPSIS in evaluating the performance of listed firms, particularly in the context of the Pakistan Stock Exchange. Most previous studies (e.g., Wanke et al., 2016; Bulgurcu, 2018) have assigned equal weights to financial ratios, ignoring the varying significance of each ratio. Additionally, there is a lack of research validating MCDM-based rankings with actual stock performance data. This study fills these gaps by using entropy for weight derivation, applying the TOPSIS method, and validating the rankings with real market returns.

2. LITERATURE REVIEW

The literature reports several studies applying different multi-criteria decision-making techniques for the evaluation of firms using their financial ratios. For instance, Aduba (2022) proposed a framework to evaluate the financial performance of Japanese firms using fuzzy analytical hierarchical process (FAHP) and TOPSIS method. The study used normalized multivariate linear correlation between performance criteria. Similarly, (Bae et al., 2021) used FAHP and TOPSIS approach to compare the analysis of several airlines using financial ratios. In another study, Ksenija et al., (2017) used FAHP and TOPSIS to assess the efficiency of insurance companies in Serbia. The study used five key financial indicators of insurance sector companies for the period from 2007 to 2014. The study found weights using FAHP and then used TOPSIS for selection criteria ranking. (Shaverdi et al., 2024) used financial ratios to develop a framework for ranking petrochemical industry firms in Iran. Likewise, they used Fuzzy AHP for weight determination and TOPSIS method to compare the firms. In addition to FAHP and TOPSIS methods, the other alternative methods of multi-criteria decision making have also been reported. For instance, Türegün (2021) used TOPSIS and VIKOR method supported by entropy approach to measure the financial performance of tourism sector firms listed on Borsa stock exchange Istanbul. The study took data of the period of 2018 to 2020. (Aras et al., 2018) used entropy method followed by TOPSIS method to evaluate the relative performance of financial sector firms. The study reported that Empirical findings indicate that the average performances of continuously operating intermediary institutions during the concerned period are above the average performance levels of all intermediaries.

There are several studies that have been done to measure relative performance of intermediary institutions as well. For example, Tunay and Akhisar (2015) assessed the performance of private banks. The study also employed TOPSIS scores for the the period of 2009 to 2013. They reported higher the capital adequacy ratio, the higher the level of protection available to depositors. Likewise, Başçı (2016) assessed the performance and ranked Turkish private banks using AHP and TOPSIS, the branch capability was also considered a significant factor in the study. The study reported several ways of cost reduction. Okay & Köse (2015) assessed the same for five firms considering ten financial ratios. The study employed TOPSIS methodology taking the data from 2011 to 2014. The study reported the profitability ratios being more important for financial performance of a firm. Furthermore, Tarihi et al., (2017) also reported a comparative study on five brokerage houses using financial ratios from 2014 to

2015. They employed the method of ELECTRE, ORESTE and TOPSIS. Each method resulted almost same ranking for 2014 but different for 2015.

2.1 Hypotheses

1. **H₁** : The entropy-based weighting method significantly improves the evaluation of firm performance compared to the equal-weight method.
2. **H₂** : There is a significant positive correlation between the firm rankings generated by the proposed hybrid model and the actual stock return rankings.
3. **H₃** : The use of complex numbers and MATLAB significantly improves the handling of negative values in entropy calculations.\

3. METHODOLOGY

The TOPSIS method was proposed by Hwang and Yoon in 1981 (Chen & Tsao, 2008). They introduced the concept of compromised solution. The compromised solution is choosing the best alternative nearest to the positive ideal solution and at the same time farthest from the negative ideal solution. The closeness in TOPSIS is measured using minimum Euclidean distance.

The preliminary step for using TOPSIS is determination of weights of alternative criteria using entropy approach. The actual data containing this information in matrix form is called decision matrix. The steps for finding entropy weights are given as under.

3.1 Steps for finding criteria weights using entropy

The first step is to define a decision matrix D with several alternatives in rows and their selection criteria in columns.

We then calculate the normalized matrix N by dividing each of elements in the decision matrix D by the sum of its column. The formula for $N_{i,j} = \frac{x_{i,j}}{\sum_{i=1}^{m} x_{i,j}}$ where x shows the individual elements of D and m is the number of alternatives.

The next step is to compute the entropy vector measure. The entropy vector will have dimensions $1 \times r$ where r is the number of selection criteria. The formula to find the entropy vector is $E_j = -k \sum_{i=1}^{m} N_{i,j} \times \ln(N_{i,j})$ where $k = \frac{1}{\ln m}$. In this formula, N shows the elements of normalized matrix and m is the number of alternatives.

The final step is to create the weight vector W based on Entropy vector. The method to find this is to divide $1 - E_j$ by its sum. This vector will show the respective weights of each selection criteria and their sum must be equal to 1.

3.2 Steps for using TOPSIS method

The preliminary of using TOPSIS method is to have the weights of each selection criteria and the attribute of each selection criteria as either beneficial or non-beneficial. An example of beneficial criteria could be sales and non-beneficial could be cost as the higher value of sales and lower value of cost are preferred.

In the first step we find the normalized matrix using vector normalization method. In this method each element is divided by square root of squared sum of each column. This method transforms each element into non-dimensional attribute. The formula for normalized matrix is given by $N_{i,j} = \frac{x_{i,j}}{\sqrt{\sum_{i=1}^{m} x_{i,j}^2}}$. where x_{ij} is the individual element of the decision matrix D.

The second step after creating a non-dimensional matrix is to create a weighted normalized matrix V. The formula for this matrix is given by $V_{ij} = W_j \times N_{ij}$. Here we multiply the respective weights of each selection criteria by the individual value of normalized matrix.

The next step is to find the positive best P and negative best N for each criterion. In the case of beneficial criterion, the positive best is the maximum criterion for each alternative whereas the negative best is the minimum value of each criterion for each alternative. The opposite rule applies for the non-beneficial criterion.

After finding the positive best and negative best vectors, we find the Euclidian distance for each element of weighted normalized matrix V_{ij} from positive best and negative best in S_i^+ and S_i^- matrices using the Euclidian distance formula $S_i^+ = \sqrt{\sum_{i=1}^{n} \left(V_{ij} - P_j^+\right)^2}$ and $S_i^- = \sqrt{\sum_{i=1}^{n} \left(V_{ij} - P_j^-\right)^2}$.

The Final step is to calculate the performance score vector P_i using the formula $P_i = \frac{S_i^-}{S_i^- + S_i^+}$ and rank the results of P_i.

4. DATA ANALYSIS AND RESULTS

In this study we use TOPSIS approach to obtain the ranking of the ten selected companies listed on Pakistan stock exchange for 2019 to 2024 based on their financial ratios. Following the study by Tey et al., (2019) we use 12 financial ratios to measure the ranking of the firms. The ratios are current ratio, quick ratio, cash ratio, total asset turnover, returns of assets (ROA), returns on equity (ROE), sales turnover, fixed assets turnover, debt to equity ratio, debt to total assets ratio and net profit margin. We found these ratios from the official website of State Bank of Pakistan. In the conventional TOPSIS approach, the companies are considered as alternatives whereas the financial ratios are considered as selection criteria. The elements of matrix containing m alternatives with j criteria can be represented as $(x_{i,j})_{m \times n}$. Since we have 10 alternatives firms with 12 alternative criteria, out decision matrix will be of order 10×12 for each year.

4.1 Determination of weight using entropy

In several studies based on financial ratios, equal weights have been assigned to each ratio. This is not practically valid. For example, the current ratio and cash ratio are not equally important for a decision maker while ranking the companies. Similarly, the debt-to-equity ratio and sales turnover are entirely different criteria to be considered as equally important. To address this gap, we use entropy method of weight determination. The entropy method uses

a systematic probability approach to assign weight to each criterion. The part of this section comprises finding the weights of each criterion using entropy method. This study uses 12 financial ratios of 10 selected companies listed at Pakistan Stock Exchange. The actual data containing this information in matrix form is called decision matrix. We only show working for the year 2020 for the purpose of brevity. The other years also follow the same methodology. For the year 2020, the decision matrix D is given in table 4.1 below.

Table 4.1 Decision matrix containing selection criteria-based o financial ratios and alternative firms

	L1	L2	A1	P1	P2	L3	A2	A3	A4	S1	S2	P3
C1	1.5230	1.0612	4.0254	15.3839	34.6219	0.5124	3.6141	116.8556	23.6340	1.3546	0.6408	3.8217
C2	1.2308	0.9453	1.0986	4.0076	8.2566	0.6330	0.9998	1286.6496	1.7090	1.0740	0.5690	3.6478
C3	0.3723	0.0908	1.0901	1.8025	22.6191	0.0044	0.8312	109.7300	1.0350	6.0321	1.1250	1.6535
C4	1.3637	0.5276	1.8679	13.9870	20.0737	0.1660	1.7504	1442.3056	2.7811	0.4164	0.3137	7.4881
C5	2.2218	1.6954	0.4813	17.0744	31.3974	1.3759	0.4732	28.6551	0.7850	0.8303	0.4614	35.4778
C6	2.8904	2.6485	0.3921	11.9577	17.4588	0.1131	0.3608	44.0525	0.6929	0.5008	0.3627	30.4981
C7	0.6477	0.2480	2.7611	4.1824	40.2476	0.0341	2.7000	1518.2263	5.6495	7.3595	0.9003	1.5147
C8	1.3093	0.9261	2.3903	4.9610	18.7477	0.0147	2.2376	465.8628	36.7681	2.8157	0.7883	2.0755
C9	0.8153	0.1990	4.1484	7.8196	29.8735	0.0906	4.3413	3010.5810	10.8423	2.8138	0.7050	1.8849
C10	0.9409	0.4729	3.3725	2.5706	16.4969	0.2166	2.9937	1542.1484	10.9515	5.1041	0.9420	0.7622

The normalized matrix for 2020 is given in table 4.2 below

Table 4.2 Normalized matrix

	L1	L2	A1	P1	P2	L3	A2	A3	A4	S1	S2	P3
C1	0.1144	0.1204	0.1861	0.1837	0.1444	0.1621	0.1780	0.0122	0.2492	0.0479	0.0941	0.0430
C2	0.0924	0.1072	0.0508	0.0479	0.0344	0.2003	0.0492	0.1345	0.0180	0.0379	0.0836	0.0411
C3	0.0280	0.0103	0.0504	0.0215	0.0943	0.0014	0.0409	0.0115	0.0109	0.2131	0.1652	0.0186
C4	0.1024	0.0599	0.0864	0.1670	0.0837	0.0525	0.0862	0.1508	0.0293	0.0147	0.0461	0.0843
C5	0.1669	0.1923	0.0223	0.2039	0.1309	0.4353	0.0233	0.0030	0.0083	0.0293	0.0678	0.3994
C6	0.2171	0.3005	0.0181	0.1428	0.0728	0.0358	0.0178	0.0046	0.0073	0.0177	0.0533	0.3434
C7	0.0486	0.0281	0.1277	0.0499	0.1678	0.0108	0.1330	0.1587	0.0596	0.2600	0.1322	0.0171
C8	0.0983	0.1051	0.1105	0.0592	0.0782	0.0046	0.1102	0.0487	0.3877	0.0995	0.1158	0.0234
C9	0.0612	0.0226	0.1918	0.0934	0.1246	0.0287	0.2138	0.3147	0.1143	0.0994	0.1036	0.0212
C10	0.0707	0.0536	0.1559	0.0307	0.0688	0.0685	0.1475	0.1612	0.1155	0.1803	0.1384	0.0086

The next step is to compute the entropy vector measure. The entropy vector will have dimensions $1 \times r$ where r is the number of ratios. In this case it will be 1×12 vector. The results for the entropy support vector E calculated as $N_{i,j} \times \ln(N_{i,j})$ are shown in table 4.3.

Table 4.3 Entropy Support Matrix

-0.2480	-0.2549	-0.3129	-0.3113	-0.2794	-0.2949	-0.3072	-0.0538	-0.3463	-0.1455	-0.2224	-0.1354
-0.2201	-0.2394	-0.1514	-0.1455	-0.1160	-0.3221	-0.1483	-0.2698	-0.0724	-0.1241	-0.2074	-0.1311
-0.1000	-0.0471	-0.1506	-0.0826	-0.2227	-0.0092	-0.1308	-0.0513	-0.0493	-0.3295	-0.2975	-0.0742
-0.2334	-0.1685	-0.2115	-0.2989	-0.2076	-0.1548	-0.2113	-0.2853	-0.1035	-0.0621	-0.1418	-0.2085
-0.2988	-0.3171	-0.0847	-0.3242	-0.2662	-0.3620	-0.0876	-0.0174	-0.0397	-0.1035	-0.1824	-0.3666
-0.3316	-0.3613	-0.0727	-0.2779	-0.1908	-0.1191	-0.0716	-0.0248	-0.0359	-0.0714	-0.1562	-0.3670
-0.1471	-0.1005	-0.2628	-0.1497	-0.2996	-0.0488	-0.2683	-0.2921	-0.1680	-0.3503	-0.2675	-0.0694
-0.2281	-0.2367	-0.2434	-0.1674	-0.1993	-0.0249	-0.2431	-0.1472	-0.3674	-0.2296	-0.2496	-0.0878
-0.1710	-0.0856	-0.3167	-0.2214	-0.2595	-0.1018	-0.3299	-0.3638	-0.2479	-0.2295	-0.2348	-0.0818
-0.1872	-0.1569	-0.2898	-0.1069	-0.1841	-0.1837	-0.2823	-0.2942	-0.2493	-0.3089	-0.2737	-0.0408

Multiplying the sum of each column by $-k$ gives the following vector E

$$E = [0.9404 \quad 0.8547 \quad 0.9105 \quad 0.9058 \quad 0.9664 \quad 0.7042 \quad 0.9035 \quad 0.7816 \quad 0.7294 \quad 0.8488 \quad 0.9700 \quad 0.6786]$$

The next step is to create the weight vector W based on Entropy vector. The method to find this is to divide $1 - E_j$ by its sum. The results for the weight vector W are given as under

$$W_j = [0.0330 \quad 0.0804 \quad 0.0496 \quad 0.0521 \quad 0.0186 \quad 0.1638 \quad 0.0534 \quad 0.1209 \quad 0.1498 \quad 0.0837 \quad 0.0166 \quad 0.1780]$$

It is worth noting that the sum of all weights must be equal to 1.

The next step was to create matrix defining beneficial and non-beneficial criteria. Since the greater value of current ratio, quick ratio, cash ratio, total assets turnover ROA, ROE, sales turnover, fixed assets turnover, total assets turnover, net profit margin is preferable, we consider them as beneficial criteria. The debt to equity and debt to total assets were considered non-beneficial criteria. Table 4.4 shows the beneficial and non-beneficial criteria ratios

Table 4.4 Beneficial and non-beneficial criterion

L1	L2	A1	P1	P2	L3	A2	A3	A4	S1	S2	P3
1	1	1	1	1	1	1	1	1	0	0	1

The final stage is to apply TOPSIS methodology. The result for the normalize matrix are shown in table 4.5.

Table 4.5 Normalized matrix for TOPSIS approach

	L1	L2	A1	P1	P2	L3	A2	A3	A4	S1	S2	P3
C1	0.3189	0.2901	0.5016	0.4875	0.4259	0.3146	0.4734	0.0278	0.5047	0.1159	0.2794	0.0799
C2	0.2577	0.2585	0.1369	0.1270	0.1016	0.3886	0.1310	0.3056	0.0365	0.0919	0.2481	0.0763
C3	0.0780	0.0248	0.1358	0.0571	0.2783	0.0027	0.1089	0.0261	0.0221	0.5162	0.4906	0.0346
C4	0.2856	0.1443	0.2328	0.4432	0.2469	0.1019	0.2293	0.3426	0.0594	0.0356	0.1368	0.1566
C5	0.4652	0.4635	0.0600	0.5410	0.3862	0.8447	0.0620	0.0068	0.0168	0.0711	0.2012	0.7420
C6	0.6052	0.7241	0.0489	0.3789	0.2148	0.0694	0.0473	0.0105	0.0148	0.0429	0.1582	0.6378
C7	0.1356	0.0678	0.3441	0.1325	0.4951	0.0209	0.3537	0.3606	0.1206	0.6298	0.3926	0.0317
C8	0.2741	0.2532	0.2979	0.1572	0.2306	0.0090	0.2931	0.1106	0.7852	0.2410	0.3437	0.0434
C9	0.1707	0.0544	0.5170	0.2478	0.3675	0.0556	0.5687	0.7150	0.2315	0.2408	0.3074	0.0394
C10	0.1970	0.1293	0.4203	0.0815	0.2029	0.1330	0.3922	0.3663	0.2339	0.4368	0.4108	0.0159

After creating a non-dimensional matrix, we create a weighted normalized matrix by formula $V_{ij} = W_j \times N_{ij}$. Table 4.6 shows the outcome of weighted normalized matrix.

Table 4.6 Weighted normalized matrix

	L1	L2	A1	P1	P2	L3	A2	A3	A4	S1	S2	P3
C1	0.0105	0.0233	0.0249	0.0254	0.0079	0.0515	0.0253	0.0034	0.0756	0.0097	0.0046	0.0142
C2	0.0085	0.0208	0.0068	0.0066	0.0019	0.0637	0.0070	0.0369	0.0055	0.0077	0.0041	0.0136
C3	0.0026	0.0020	0.0067	0.0030	0.0052	0.0004	0.0058	0.0032	0.0033	0.0432	0.0082	0.0062
C4	0.0094	0.0116	0.0115	0.0231	0.0046	0.0167	0.0123	0.0414	0.0089	0.0030	0.0023	0.0279
C5	0.0154	0.0373	0.0030	0.0282	0.0072	0.1383	0.0033	0.0008	0.0025	0.0059	0.0033	0.1320
C6	0.0200	0.0583	0.0024	0.0198	0.0040	0.0114	0.0025	0.0013	0.0022	0.0036	0.0026	0.1135
C7	0.0045	0.0055	0.0171	0.0069	0.0092	0.0034	0.0189	0.0436	0.0181	0.0527	0.0065	0.0056
C8	0.0091	0.0204	0.0148	0.0082	0.0043	0.0015	0.0157	0.0134	0.1176	0.0202	0.0057	0.0077
C9	0.0056	0.0044	0.0256	0.0129	0.0068	0.0091	0.0304	0.0864	0.0347	0.0202	0.0051	0.0070
C10	0.0065	0.0104	0.0208	0.0042	0.0038	0.0218	0.0210	0.0443	0.0350	0.0366	0.0068	0.0028

The next step is to find the positive best and negative best for each criterion. In the case of beneficial criterion, the positive best is the maximum criterion for each alternative whereas the negative best is the minimum value of each criterion for each alternative. The opposite rule applies for the non-beneficial criterion. The positive best vector P and negative best vector N are given as under

$$P_j = [0.0200 \quad 0.0583 \quad 0.0256 \quad 0.0282 \quad 0.0092 \quad 0.1383 \quad 0.0304 \quad 0.0864 \quad 0.1176 \quad 0.0030 \quad 0.0023 \quad 0.1320]$$

$$N_j = [0.0026 \quad 0.0020 \quad 0.0024 \quad 0.0030 \quad 0.0019 \quad 0.0004 \quad 0.0025 \quad 0.0008 \quad 0.0022 \quad 0.0527 \quad 0.0082 \quad 0.0028]$$

After finding the positive best and negative best vectors, we find the Euclidian distance for each element of weighted normalized matrix V_{ij} from positive best and negative best in S_i^+ and S_i^- matrices using the Euclidian distance formula. The findings of the S_i^+ and S_i^- are as follows.

$$S_i^+ = [0.1774 \quad 0.1940 \quad 0.2482 \quad 0.2058 \quad 0.1493 \quad 0.1961 \quad 0.2283 \quad 0.2053 \quad 0.2070 \quad 0.2079]$$

$$S_i^- = [0.1099 \quad 0.0889 \quad 0.0122 \quad 0.0763 \quad 0.2000 \quad 0.1363 \quad 0.0516 \quad 0.1237 \quad 0.1049 \quad 0.0667]$$

Now we calculate the performance score vector P_i using the formula $P_i = \frac{S_i^-}{S_i^- + S_i^+}$. The results for vector P are as under

$$P = [0.3825 \quad 0.3144 \quad 0.0468 \quad 0.2705 \quad 0.5725 \quad 0.4100 \quad 0.1845 \quad 0.3760 \quad 0.3363 \quad 0.2430]$$

The final step is to rank the performance scores. Table 4.7 shows the rank of the companies according to TOPSIS method for the year 2020.

Table 4.7 Final rankings of TOPSIS

Company	Rank
Attock Petroleum Ltd.	8
Attock Refinery Ltd.	5
Byco Petroleum	1
National Refinery Ltd.	4
Pakistan Oilfields Ltd.	10
Pakistan Petroleum Ltd.	9
Pakistan Refinery Ltd.	2
Pakistan State Oil Co. Ltd.	7
Shell Pakistan Ltd.	6
Hascol Petroleum Ltd.	3

We apply the same methodology for other and table 4.8 shows the results for the ranking of the firms using TOPSIS methodology for the year 2019 to 2024.

Table 4.8 TOPSIS rankings for year 2019 to 2024

Company	2019	2020	2021	2022	2023	2024
Attock Petroleum Ltd.	4	6	8	8	5	3
Attock Refinery Ltd.	9	7	5	4	4	1
Byco Petroleum	1	1	1	6	2	4
National Refinery Ltd.	5	10	4	3	6	2
Pakistan Oilfields Ltd.	2	8	10	10	8	10
Pakistan Petroleum Ltd.	3	9	9	9	7	9
Pakistan Refinery Ltd.	10	2	2	5	9	8
Pakistan State Oil Co. Ltd.	8	5	7	7	3	5
Shell Pakistan Ltd.	7	4	6	2	1	6
Hascol Petroleum Ltd.	6	3	3	1	10	7

To check the practicality of TOPSIS results for the long-term investors, we find the ranking of the same companies based on their annual stock return. The stock returns have been calculated by formula $R_{it} = \frac{P_{it}-P_{it-1}+D_{it}}{P_{it-1}} \times 100$ where i represents the individual stock price at t years, P is the price of stock and D is the dividend distributed in the respective years. The rankings of the firms based on their stock returns revealed the following results in table 4.9

Table 4.9 Ranking of firms based on their stock returns

Company	2019	2020	2021	2022	2023	2024
Attock Petroleum Ltd.	5	2	7	8	5	9
Attock Refinery Ltd.	7	8	6	2	3	10
Byco Petroleum	10	4	1	4	6	6
National Refinery Ltd.	8	10	8	3	2	7
Pakistan Oilfields Ltd.	2	3	5	10	9	4
Pakistan Petroleum Ltd.	3	5	2	9	10	2
Pakistan Refinery Ltd.	1	1	4	5	4	3
Pakistan State Oil Co. Ltd.	6	6	3	6	7	5
Shell Pakistan Ltd.	4	7	10	1	8	8
Hascol Petroleum Ltd.	9	9	9	7	1	1

Comparing the results of both TOPSIS rankings and stock returns, we could observe that mostly the rankings are close to each other and follow the same trend except for the year 2020. Figures 1 to 4 show the results of some of selected companies rankings comparison. The outbreak of

COVID-19 might be a possible reason for the anomalous deviation for 2020. However, Increasing the sample size for a longer period of time and increasing the number of firms to a larger number might improve the results in future.

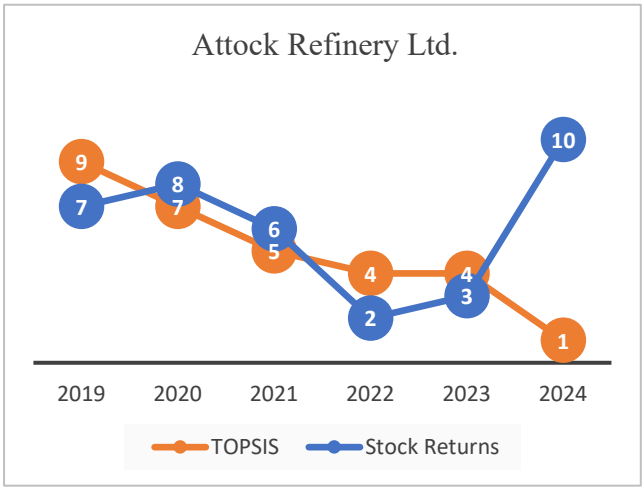


Figure 1 Comparison of TOPSIS and stock returns of firms

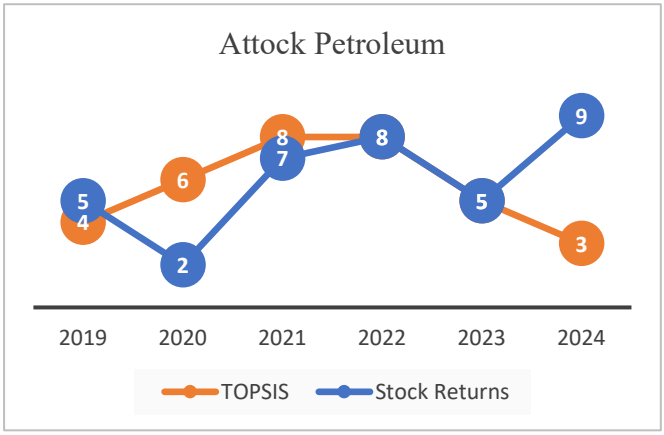


Figure 2 Comparison of TOPSIS and stock returns of firms

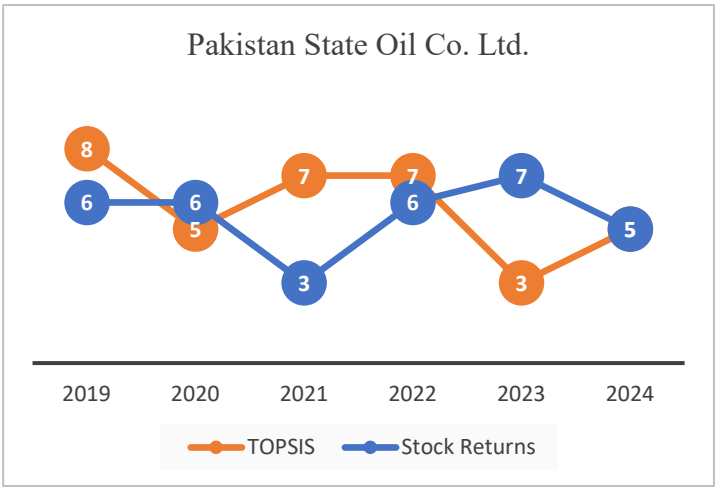


Figure 3 Comparison of TOPSIS and stock returns of firms

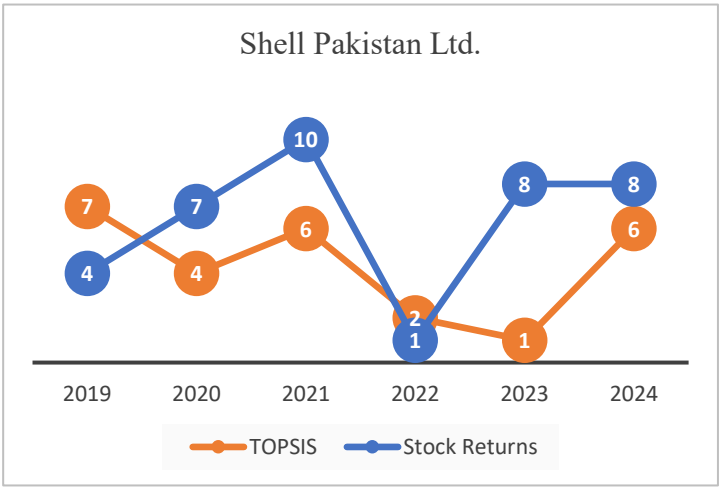


Figure 4 Comparison of TOPSIS and stock returns of firms

5. Conclusion and Recommendations

This study uses a hybrid approach of entropy and TOPSIS to rank the performance of different alternative firms for a long-term investor to select best firms for the portfolio. We used entropy method to determine the weights of selection criteria based the financial performance of firms and TOPSIS method to determine the ranking for each alternative firm based on ideal minimum (maximum) distance for the best (worst) possible criteria. Most of the studies have limitation of using equal weights of selection criteria. This study addresses this limitation by employing the entropy approach of weight determination. The sample size consisted of 10 firms from Refinery, Oil and Gas Marketing, and Oil and Gas Exploration sectors of Pakistan stock exchange and the financial data consisted of six years from 2019 to 2024. The results of the ranking of firms were compared with the annual stock returns of the firms. The results indicate that, in most cases, the ranks of our method are very close to the ranks based on annual stock returns of firms. Which indicates that this method produces effective and reliable results. However, Increasing the number of firms and lengthening the sample data period could produce better results. This study is useful for individuals and institutional investors to help them select best alternatives for their portfolio selection based on financial performance of companies.

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